

Candidate Biomarkers and Key Molecular Pathways in Anxiety Disorder and Parkinson's Disease Using Bioinformatics and Machine Learning

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Abstract—Anxiety disorder (AD) and Parkinson's disease (PD) are complex neuropsychiatric and neurodegenerative disorders that profoundly affect neurological function, overall health, and quality of life. Increasing clinical and experimental evidence suggests that anxiety symptoms often co-occur with Parkinson's disease, indicating the presence of shared molecular mechanisms and interconnected neurobiological pathways. Consequently, identifying common molecular determinants and therapeutic targets is essential for improving early diagnosis, understanding disease progression, and developing effective treatment strategies. In this study, an integrative systems biology framework combined with machine learning techniques was employed to pinpoint key differentially expressed genes (DEGs) from publicly available transcriptomic datasets. Functional enrichment analyses, including Gene Ontology (GO) and KEGG pathway analysis, were conducted to unveil the correlated biological activities and regulatory pathways. Protein-protein interaction (PPI) structure analysis was used to locate hub genes, while transcription factor (TF)-gene and microRNA (miRNA)-gene regulatory networks were constructed to explore upstream regulatory mechanisms. Finally, computational drug repurposing analysis was performed to identify candidate therapeutic compounds targeting both disorders. The identified molecular signatures and candidate drugs may provide potential biomarkers and therapeutic targets for early diagnosis and improved treatment strategies for AD and PD.

Index Terms—Anxiety disorder, Parkinson's disease, Differentially expressed genes, Hub genes, Protein-protein interaction network, Drug molecules.

I. INTRODUCTION

Anxiety disorder (AD) is a prevalent mental health condition characterized by excessive fear, worry, or nervousness that

disrupts daily life. The primary symptoms of AD include persistent anxiety, restlessness, difficulty concentrating, and sleep disturbances, often lasting for six months or longer [1]. These disorders encompass mental health conditions in which individuals experience persistent and intense anxiety and fear. Approximately 12% of people experience an anxiety disorder in a given year, and between 5% and 30% are affected at some point during their lifetime [2]. AD cause significant distress, interfere with routine activities, and may lead to serious complications, including disability and premature death [3]. Globally, nearly 13.4% of individuals are affected by mental health conditions, with anxiety disorders accounting for approximately 6.5% of the population [4], [9]. These disorders are frequently observed in patients with PD, a serious neurodegenerative disorder caused by the progressive loss of dopamine-producing neurons in the substantia nigra [5]. PD is recognized as the second most prevalent neurodegenerative disorder, affecting nearly seven million individuals worldwide [6]. Currently, Parkinson's disease remains incurable. As the disease progresses, patients experience increasing severity of motor and non-motor symptoms, which significantly impair quality of life, particularly among older individuals [7]. Recent studies indicate that PD results from the interaction of both genetic and environmental factors [8]. Several genes, including *LRK2* [10] and *VPS35*, have been linked to susceptibility to PD; however, the precise molecular mechanisms underlying the disease remain unclear. Consequently, knowledge regarding the key genes involved in PD progression remains limited. Anxiety disorders affect up to 18% of the population during their lifetime, whereas PD occurs in approximately 1–2% of individuals over 60 years of age [14]–[16].

Therefore, identifying novel genes associated with the growth and continuation of PD is essential. The pathophysiology of AD involves dysregulation of neurotransmitters, including serotonin, GABA, and norepinephrine, hyperactivity of the hypothalamic–pituitary–adrenal axis, and structural as well as functional changes within the amygdala, hippocampus, and prefrontal cortex [14]. In contrast, PD pathology is characterized by dopaminergic neuronal loss in the substantia nigra, α -synuclein accumulation, oxidative stress, mitochondrial dysfunction, and neuroinflammation [17], [18]. Early longitudinal studies have indicated that anxiety symptoms may precede the onset of motor symptoms in PD, suggesting a potential shared neurobiological basis between these disorders [12], [19]. Furthermore, chronic anxiety has been associated with a higher risk of developing PD, accelerated motor deterioration, and increased cognitive decline [13], [20].

In this study, we performed a systematic bioinformatics analysis to identify key molecular markers associated with both AD and PD. Gene expression data from the GSE61672 and GSE6613 datasets were analyzed, leading to the identification of 22 overlapping differentially expressed genes (DEGs). Further investigation using Enrichr and KEGG pathway enrichment analyses highlighted inflammatory responses and neurotransmitter signaling as critical biological processes involved in disease progression. Several computational studies have also identified potential drug compounds targeting these DEGs, including hydrogen peroxide, chlorzoxazone, cloperastine, danazol, cylindrospermopsin, tretinoin, tert-butyl hydroperoxide, mesalazine, and pingyangmycin, associated with molecular targets such as *TXNRD2*, *CYP1B1*, *FBXO9*, *UPP1*, *DHX40*, *OSBPL8*, *RBM15*, and *PAPOLA*. Our findings reveal common pharmacological targets that warrant further investigation, potentially contributing to improved early diagnosis, outcome prediction, and the development of more effective therapeutic strategies. The primary objective of this study is to identify biomarkers associated with disease risk, advance preventive healthcare strategies, and enhance therapeutic interventions. By applying the Fisher score method to identify key genes and integrating computational drug discovery approaches, this study provides a foundation for translational research in AD and PD, highlighting novel molecular targets and potential therapeutic strategies for these complex disorders.

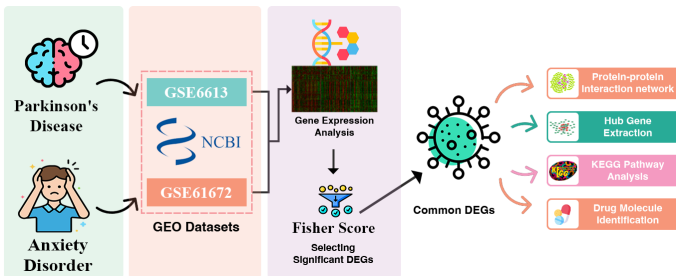


Fig. 1. The figure illustrates the investigation of shared molecular pathways and therapeutic targets in AD and PD, highlighting key DEGs identified using the Fisher Score method.

TABLE I
SUMMARY OF THE DATASETS

Dataset Name	Total Samples	Control Samples	Test Samples	Platform
GSE6613	105	55	50	GPL96
GSE61672	546	390	156	GPL10558

II. METHODOLOGY

A. Dataset description

This study utilized two publicly available microarray datasets obtained from the GEO repository: GSE61672 and GSE6613. The GSE61672 dataset has been extensively used in studies investigating anxiety disorders, while GSE6613 has been widely employed as a resource for Parkinson's disease research. The analyses were conducted using the GPL10558 and GPL96 platforms, both of which provide comprehensive gene expression coverage for transcriptomic studies. A detailed summary of all datasets used in this study is presented in Table I for clear and convenient reference.

B. Data preprocessing and DEG selection

The analysis in the present study was initiated by identifying DEGs from the GSE61672 and GSE6613 datasets. A gene is considered differentially expressed when its expression level exhibits statistically significant changes between experimental conditions. The microarray datasets GSE61672 and GSE6613 were analyzed using the limma package within the R statistical computing environment [21]. Prior to analysis, the datasets were preprocessed, and genes were filtered using an adjusted p -value threshold of less than 0.15 to determine statistical significance.

C. DEGs selection using machine learning

To identify the key DEGs shared between both datasets, the Fisher score, a supervised feature selection method, was applied. This method evaluates each feature by computing the ratio of inter-class variance to intra-class variance, thereby highlighting features that exhibit substantial variation between classes while remaining relatively consistent within each class. Using this approach, the 800 most relevant and discriminative DEGs were selected from the datasets associated with AD and PD. Subsequently, the overlapping DEGs from both datasets, identified based on Fisher score ranking, were further analyzed to reveal potential shared molecular mechanisms between AD and PD.

D. PPI network analysis and identification of hub genes

STRING platform integrates diverse sources of information on functional protein–protein interactions, providing a deeper understanding of the complexity of biological systems. To construct the PPI network, the recognized DEGs were inspected using the STRING platform. PPI networks offer critical insights into protein functions and play a central role in network biology by enabling the graphical representation of complex biological processes. In this study, PPI networks were constructed using NetworkAnalyst [23], which integrates

physical interaction data among DEG-encoded proteins obtained from the IMEx database [24]. Hub proteins, characterized by their high interaction connectivity within the network, were subsequently identified through PPI network analysis. Furthermore, Cytoscape was utilized for visualization and additional network analysis, while the *cytoHubba* plugin was applied to identify the top 10 hub proteins based on the Degree centrality metric.

E. Gene regulatory interaction network

The interactions between transcription factors (TFs) and the identified consistently regulated DEGs were investigated to evaluate TF-mediated regulation of biological processes, with particular emphasis on their influence on functional pathways and gene expression patterns. The previously identified concordant genes were analyzed for TF–gene interactions using the NetworkAnalyst platform. The RegNetwork database [22] was utilized to identify coregulatory interactions between microRNAs (miRNAs) and TFs, enabling the classification of regulatory elements affecting DEGs at both transcriptional and post-transcriptional levels. Furthermore, the TF–miRNA interaction network was analyzed and visualized using the NetworkAnalyst platform.

F. Gene ontology and pathway analysis

Enrichment analysis evaluates groups of genes associated with various biological functions and specific genomic regions. Gene Ontology (GO) terms are commonly used for assigning functional annotations to gene products, categorizing them into biological processes, cellular components, and molecular functions [20]. To elucidate the molecular mechanisms associated with both CVD and PE, a comprehensive systems biology analysis of DEGs was conducted using curated pathway databases, including KEGG, WikiPathways, and Reactome [21]. This integrative analysis highlighted significant cellular functions, molecular activities, and biological pathways shared between the two disorders. The findings provide valuable insights into potential therapeutic targets and may support the development of effective therapeutic strategies in future studies.

G. Drug signature Identification

Predicting candidate drug molecules represents an essential component of this study and constitutes a major research objective. Using DEGs with similar expression profiles, the Enrichr platform utilizes the DSigDB database to identify potential therapeutic compounds. DSigDB serves as a comprehensive repository for predicting drug candidates associated with DEGs, providing an extensive collection of drug–gene interaction signatures. Through transcriptome profile analysis and the Enrichr platform, Several candidate therapeutic drugs for AD and PD were identified through the DSigDB database, which contains approximately 22,527 gene sets. Furthermore, the Enrichr platform enables efficient access to this database, offering extensive gene sets and supporting the generation of interactive visualizations for enrichment analysis results.

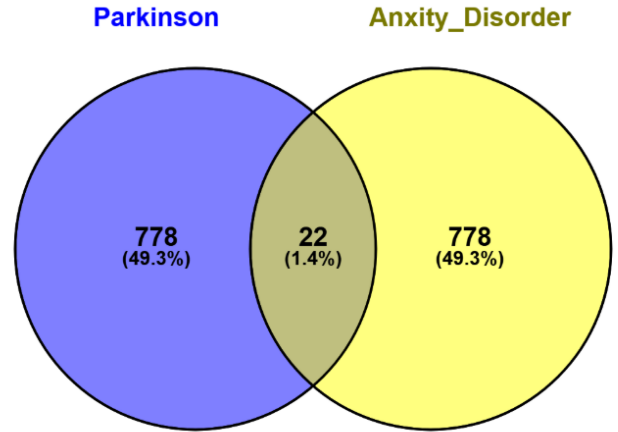


Fig. 2. Venn diagram illustrating the common DEGs between AD and PD, highlighting the 22 most significant shared genes.

III. RESULTS

A. Identification of DEGs and common genetic signatures between AD and PD

To identify differentially expressed genes (DEGs) and uncover shared genetic signatures associated with both AD and PD, two publicly available microarray datasets, GSE61672 and GSE6613, were analyzed. The GSE61672 dataset initially contained 12,603 genes. After removing missing values and applying an modified p -value < 0.15 to retain a sufficient number of potentially relevant DEGs, a total of 1,664 DEGs were uncovered. The expression profiles of the 22 most significant genes are illustrated in Fig. 2, enabling visualization of their distinctive expression patterns. The GSE6613 dataset originally comprised 22,283 genes. Following the removal of missing values and the application of a p -value threshold of < 0.15 , a total of 4,827 DEGs were selected for further analysis.

B. Extraction of ML-based Significant DEGs

Significant DEGs were identified using machine learning (ML) approaches through the Fisher score feature selection method. The curated DEGs obtained from the two datasets were subjected to advanced computational analysis to reduce data redundancy and enable systematic ranking of features built on their biological significance. By applying the Fisher score method, the top 800 most significant DEGs were selected from each dataset, resulting in a refined set of genetic markers with strong predictive potential. A comparative analysis of the selected genes revealed 22 DEGs that were common to both AD and PD, suggesting that these neurological disorders may share underlying molecular mechanisms. The genes selected using the machine learning approach are illustrated in Fig. 2, along with a Venn diagram representing their shared distribution across the two datasets.

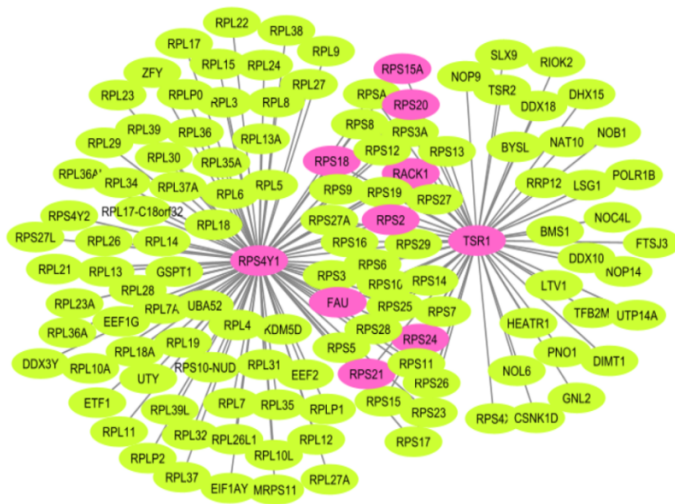


Fig. 3. The PPI network was generated using the 22 shared DEGs. Purple nodes represent the hub genes, while olive green nodes indicate the interactions between the leading genes and other genes. The PPI network contains 124 nodes, 154 edges, and 2 seed genes.

C. PPI Network Analysis

To enhance the scope and precision of the analysis, common DEGs were first identified and subsequently analyzed using NetworkAnalyst to build a PPI network. This framework integrated the IMEx Interactome database with the NetworkAnalyst platform, leveraging a comprehensive dataset curated from the literature and accessible through InnateDB. As a result, the PPI network consisted of 124 nodes and 154 edges, with the two seed genes corresponding to the DEGs shared between AD and PD. As illustrated in Fig. 3, the PPI network highlights the complex interactions among these molecular entities, underscoring their potential relevance in disease pathophysiology.

TABLE II
TOPOLOGICAL ANALYSIS OF PRINCIPAL HUB GENES IN THE PPI NETWORK

Gene Name	Degree	Closeness Centrality	Stress	Betweenness Centrality
RPS4Y1	8	0.9000	56	0.3889
TSR1	8	0.9000	56	0.3889
RPS18	2	0.5625	2	0.0035
RPS24	2	0.5625	2	0.0035
RPS15A	2	0.5625	2	0.0035
RACK1	2	0.5625	2	0.0035
RPS20	2	0.5625	2	0.0035
RPS2	2	0.5625	2	0.0035
FAU	2	0.5625	2	0.0035
RPS21	2	0.5625	2	0.0035

D. Detection of key hub genes

PPI network analysis using *CytoHubba* identified ten hub genes that act as key biological components within the network. *RPS4Y1* achieved the highest score of 95.0, designating

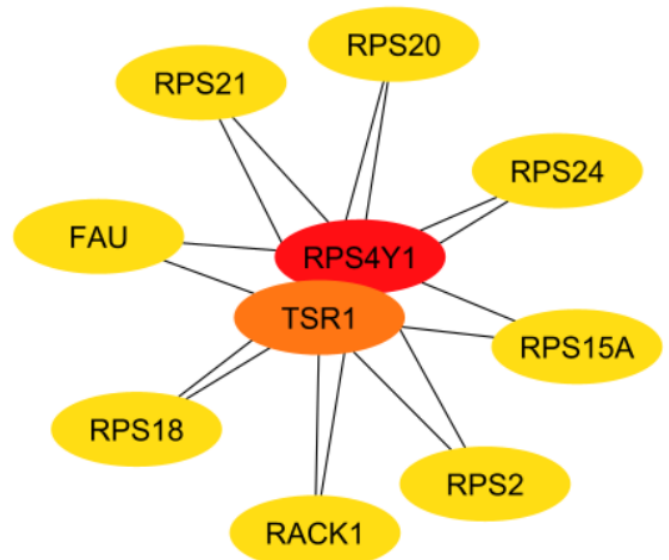


Fig. 4. Top 10 hub genes identified based on degree analysis, with RPS4Y1 and TSR1 showing the highest significance.

it as the most highly connected node and indicating substantial potential for protein–protein interactions. *TSR1* was the second most connected node, with a score of 59, highlighting its significant network relevance. As illustrated in Fig. 4, the interactions among these central hub genes are visually represented, demonstrating their network connectivity and likely functional associations. Key topological metrics for the ten hub genes are provided in Table II, calculated using the NetworkAnalyzer functionality within Cytoscape.

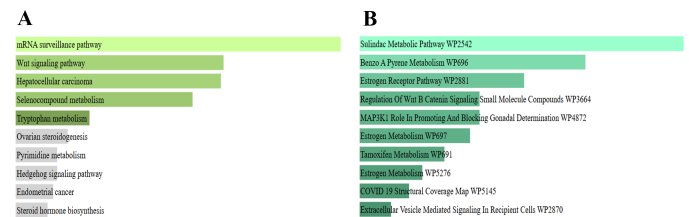


Fig. 5. Functional enrichment analysis of DEGs in AD and PD. Bar graphs of KEGG (A) and WikiPathways (B) illustrate the major pathways shared by both conditions.

E. Analyze GO and pathways based on the prevalent DEGs

Several databases were utilized to perform functional enrichment analysis of the DEGs associated with AD and PD to gain deeper insights into the underlying molecular pathways. Significant molecular pathways shared between the two conditions were identified through KEGG (Fig. 5A) and WikiPathways (Fig. 5B), which are illustrated using bar charts. Furthermore, Gene Ontology (GO) enrichment analysis was conducted, and the results are presented as bar plots in Fig. 6 (A–C). To elucidate the functional roles, subcellular localization, and biological activities of the identified DEGs, enriched terms were categorized into biological processes

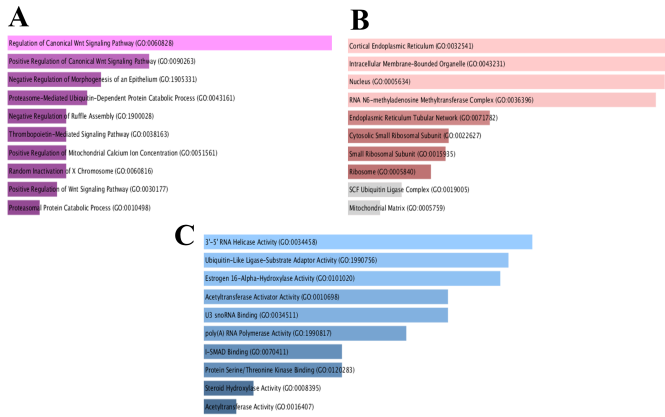


Fig. 6. Bar plots representing GO analysis of significant biological processes (A), cellular components (B), and molecular functions (C).

(Fig. 6A), cellular components (Fig. 6B), and molecular functions (Fig. 6C).

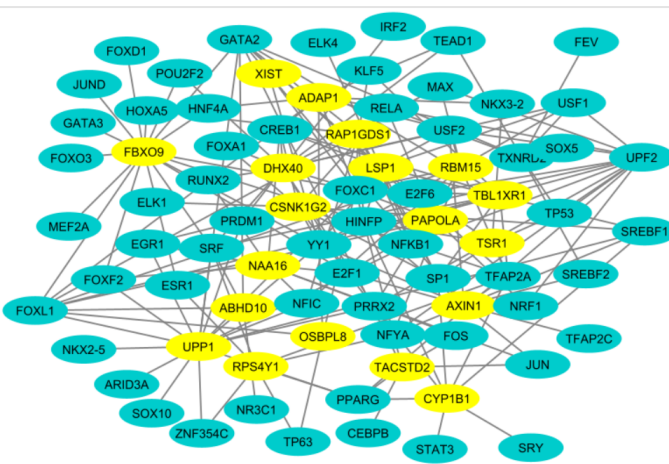


Fig. 7. Network of TF-gene interactions with common DEGs, consisting of 79 nodes and 192 edges..

F. TF-gene interactions

The TF–gene interaction network was constructed using the NetworkAnalyst platform. Figure 7 presents the transcription factor–gene associations identified for the 22 concordant DEGs. The constructed network comprises 79 nodes and 192 edges. A total of 19 TFs were identified in the TF–gene interaction network. Within the TF–gene network, *FBXO9* and *UPP1* are regulated by 17 TFs, *DHX40* by 14 TFs, and *LSP1* and *UPF2* by 13 TFs. Several other genes are regulated by 10, 8, 7, or 5 TFs, highlighting their central roles within the network, as illustrated in Fig. 7.

G. TF-miRNA coregulatory network

The TF-miRNA regulatory network was established by analyzing the regulatory interactions among transcription factors, microRNAs, and DEGs. The resulting network consists of 89 nodes and 114 edges, highlighting the complex regulatory



Fig. 8. The TF-miRNA coregulatory network consists of 89 nodes and 114 edges, depicting the relationships among TFs, microRNAs, and corresponding DEGs.

relationships among TFs, miRNAs, and DEGs. The network primarily illustrates the interactions between transcription factors and microRNAs in regulating the expression of coordinately controlled DEGs, suggesting their significant regulatory roles. As shown in Fig. 8, the network provides a visual overview of the diverse regulatory interactions that influence gene expression.

H. Therapeutic Drug Identification

Ten potential therapeutic compounds targeting genes commonly dysregulated in both AD and PD were identified using the DSigDB database via the Enrichr platform. The predicted drug candidates are summarized in Table III. Among them, clindamycin showed the strongest association (p-value: 2.1267×10^{-4} ; adjusted p-value: 0.0686), interacting with key genes such as DHX40, OSBPL8, ABHD10, RBM15, RAP1GDS1, PAPOLA, CYP1B1, and FBXO9. Additionally, several notable compounds, including hydrogen peroxide, chlorzoxazone, cloperastine, danazol, cylindrospermopsin, and tretinoin, were linked to important molecular targets. These findings highlight shared pharmacological targets that may warrant further investigation to support improved prevention and therapeutic strategies for AD and PD.

IV. CONCLUSION

This study provides important insights into the molecular associations between AD and PD. Using the Fisher score method, we identified 22 common DEGs by integrating differential gene expression analysis, PPI networks, pathway enrichment, and regulatory network analysis. Key hub genes were identified through PPI network construction, while functional enrichment analysis revealed significant biological processes, molecular functions, cellular components, and signaling pathways associated with these DEGs. Furthermore, regulatory network analysis illustrated the complex interactions among transcription factors, microRNAs, and the identified genes.

TABLE III
PREDICTED DRUG CANDIDATES BASED ON SHARED DEGS.

Name of Drugs	P-value	Adjusted P-value	Genes
clindamycin HL60 DOWN	2.1267E-4	0.0686	DHX40; OSBPL8; ABHD10; RBM15; RAP1GDS1; PAPOLA; CYP1B1; FBXO9
hydrogen peroxide CTD 00006118	2.4882E-4	0.0686	UPF2; NARS; OSBPL8; RBM15; TBL1XR1; TACSTD2; CYP1B1; UPP1; CSNK1G2; FBXO9
chlorthalidone HL60 DOWN	3.2355E-4	0.0686	DHX40; ABHD10; RBM15; TBL1XR1; PAPOLA; CYP1B1; FBXO9
AH 6809 CTD 00002290	5.9738E-4	0.0686	AXIN1; CYP1B1
AH 23848 CTD 00002080	5.9738E-4	0.0686	AXIN1; CYP1B1
eugenol CTD 00005949	7.0460E-4	0.0686	TXNRD2; CYP1B1; FBXO9
cloperastine PC3 DOWN	7.5151E-4	0.0686	PAPOLA; CYP1B1
danazol HL60 DOWN	0.001719	0.1235	UPF2; DHX40; OSBPL8; RBM15; CYP1B1; FBXO9
Cylindrospermopsin CTD 00003136	0.001739	0.1235	TXNRD2; CYP1B1; TSR1; UPP1
tretinoin HL60 UP	0.002246	0.1300	CYP1B1; LSP1; UPP1

Computational drug screening also suggested several potential therapeutic compounds capable of targeting the shared molecular mechanisms underlying both AD and PD. Overall, these findings highlight the importance of investigating overlapping molecular pathways to enhance early diagnosis, prognostic assessment, and the development of effective therapeutic strategies. In the future, we plan to expand our analysis to larger and more diverse datasets, including RNA-seq data, and incorporate advanced ML approaches. Further experimental validation and clinical investigations are crucial to verify the stability, efficacy, and translational potential of the identified targets and candidate drugs.

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